



Integrating machine learning with single-step GBLUP for enhanced genomic prediction in white spruce

Eduardo Pablo Cappa^{1,2} · Charles Chen³ · Andy Benowicz⁴ · Barb R. Thomas⁵ · Yousry A. El-Kassaby⁶

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Abstract

We evaluated the predictive performance of machine learning-augmented single- and multi-trait single-step genomic best linear prediction (ssGBLUP) models for 30 complex traits spanning productivity, chemical defense, and climate adaptability in white spruce (*Picea glauca*) from Alberta, Canada, genotyped at 211,061 SNPs. We extend a novel integration of the ssGBLUP with non-linear kernels for forest tree improvement. Using the conventional ssGBLUP model with VanRaden's linear genomic relationship matrix as baseline, we compared two non-linear extensions: an averaged Gaussian kernel (GK) with trait-specific bandwidths and an arc-cosine kernel (AK) with 1–20 layers (depth). To relate prediction gains to genetic architecture, we also fitted an extended GBLUP model explicitly including additive, dominance, and epistatic (additive × additive and additive × dominance) effects. Additive effects predominated in 21 traits, dominance contributed > 50% of variance in several chemical-defense traits, and epistasis explained ~25% in key physiological traits. Predictive performance varied by kernel type and trait architecture. AK achieved the highest accuracy in 29 traits, improving prediction by up to 27% (predictive differences up to ~0.10), particularly for traits with more complex genetic architectures. GK yielded modest gains (< 7%; predictive differences < 0.03), while linear G kernel remained competitive for core productivity traits. In multi-trait models, non-linear kernels performed similar to the linear alternatives. Notably, low-heritability traits such as drought resistance improved from 0.446–0.517 to 0.461–0.547 accuracy with multi-trait integration. These findings suggest that matching kernel complexity to genetic architecture may enhance genomic prediction and highlight the potential of non-linear kernels, particularly AK, in forest tree breeding.

Keywords Machine-learning · Predictive ability · Genomic prediction · Single- and multi-trait models · Tree breeding

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✉ Eduardo Pablo Cappa
cappa.eduardo@inta.gob.ar

- ¹ Instituto de Recursos Biológicos, Centro de Investigación en Recursos Naturales, Instituto Nacional de Tecnología Agropecuaria (INTA), De Los Reseros y Dr. Nicolás Repetto s/n, Hurlingham, Buenos Aires 1686, Argentina
- ² Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Buenos Aires, Argentina
- ³ Department of Biochemistry and Molecular Biology, Oklahoma State University, Stillwater, OK 74078, USA
- ⁴ Forest Stewardship and Trade Branch, Alberta Forestry and Parks, Edmonton, AB T6H 5T6, Canada
- ⁵ Department of Renewable Resources, University of Alberta, 442 Earth Sciences Bldg, Edmonton, AB T6G 2E3, Canada
- ⁶ Department of Forest and Conservation Sciences, Faculty of Forestry and Environmental Stewardship, University of British Columbia, Vancouver, BC V6T 1Z4, Canada

Introduction

Conventional genomic prediction (GP), initially introduced by (Meuwissen et al. 2001), typically relies on parametric linear models such as genomic best linear unbiased prediction (GBLUP), which assumes additive marker effects and a linear mapping between genotypes and phenotypes. GBLUP is based on a standard linear kernel (G) within a linear mixed model framework (Cuevas et al. 2019). This kernel corresponds to the genomic relationship *G* matrix (VanRaden 2008), computed from the shared markers, and captures the additive genetic relationships among individuals. While effective for traits predominantly controlled by additive gene action, standard linear kernels may underperform when trait architecture involves complex genetic mechanisms such as non-additive effects: dominance, and epistasis (Gianola and de los Campos 2008; Momen et al. 2018). Complex traits are often governed by multiple

genomic regions acting collectively, often through interconnected pathways, leading to the emergence of interactions (Morota and Gianola 2014). Such genetic interactions are particularly relevant in long-lived tree species, where most economically important traits are complex and polygenic (Isik 2022), potentially limiting the predictive accuracy of standard linear models in GP.

To overcome the limitations of linear GP models, machine learning (ML) methods combined with kernel-based prediction techniques have been proposed as powerful alternatives for capturing complex, cryptic, and non-linear genetic effects (Reinoso-Peláez et al. 2022), without the need to explicitly model interactions through Hadamard products of additive and/or dominance relationship matrices (Vitezica et al. 2017). One widely used kernel-based approach in GP is the semi-parametric reproducing kernel Hilbert spaces (RKHS) method (Gianola et al. 2006; Gianola and Van Kaam 2008). The RKHS models employ a kernel matrix that offers a more flexible and general representation of covariance among individuals, compared to the linear kernel (G), which is commonly used to measure molecular similarity (Reinoso-Peláez et al. 2022). Among the various kernel functions, the Gaussian kernel (GK) is one of the most applied in RKHS. The GK includes a bandwidth (or smoothing) parameter, h , which controls the rate at which the covariance between individuals decays with genetic distance. In this context, de los Campos et al. (2010) proposed a multi-kernel approach using kernel averaging to further enhance model flexibility. Extending this idea, RKHS models provide a flexible framework capable of capturing both additive and non-additive genetic effects (Alves et al. 2019), making them especially effective for predicting traits with complex genetic architectures.

Empirical evidence from both animal breeding (e.g., Tusell et al. 2014; Momen et al. 2018) and plant breeding (e.g., Perez-Rodriguez et al. 2013; Lyra et al. 2017; Morales et al. 2023) has shown that RKHS methods often outperform linear models in predicting complex traits. Using maize and wheat data, Cuevas et al. (2016, 2017, 2018) consistently found that RKHS models with GK outperformed linear GBLUP, with gains in prediction accuracy ranging from 5 to 6% on average in maize to as high as 17%, and up to 60–68% superiority for some positively correlated wheat environments. A recent meta-analysis synthesizing 72 comparisons across 34 studies and 19 predictive methods also demonstrated that RKHS-based models frequently achieved superior predictive ability (Reinoso-Peláez et al. 2022). The application of RKHS models in forest trees GP remains relatively recent but promising (Tan et al. 2017; Chen et al. 2018; Rambolarimanana et al. 2018; Arenas et al. 2021; Paludeto et al. 2021; Cappa et al. 2022a; Hu et al. 2023; Dong et al. 2024). For instance, in a lodgepole

pine (*Pinus contorta*) population of 1,490 trees, Cappa et al. (2022a) showed that RKHS models achieved higher predictive ability and lower bias than parametric GBLUP models across 13 of 15 traits. Similarly, Tan et al. (2017) reported that RKHS provided modest but consistent improvement in predictive performance for six of eight traits in two *Eucalyptus* species and their F1 hybrids.

Deep learning (DL) is a type of ML, which itself falls within the broader field of artificial intelligence (Montesinos-López O. et al. 2021). To bridge the gap between DL and kernel methods, Cho and Saul (2009) introduced the arc-cosine kernel (AK), which mathematically mimics the covariance function of infinitely wide neural networks. This approach was subsequently introduced to GP (Cuevas et al. 2019). In this context, deep kernels aim to model the genomic relationship matrix by emulating multiple hidden layers, enabling the capture of different levels of depth within the same genomic signal (Costa-Neto et al. 2020). Unlike GK, that requires careful tuning of a continuous bandwidth over the range $(0, +\infty)$, AK avoids this costly search and instead, relies on a main hyperparameter known as the “number of layers”, which is discrete and easier to optimize. Rahbar et al. (2023) further demonstrated that AK captures angular similarity and may achieve lower error or faster convergence compared to GK.

The value of incorporating non-linear kernels such as GK and AK have been seen in animal and crop plants. Working with maize and wheat data, for example, Cuevas et al. (2019) reported that these two non-linear kernels outperformed linear approaches in prediction accuracy when incorporating both molecular markers and near-infrared spectroscopy data. In maize, Costa-Neto et al. (2020) showed that both kernels yielded higher predictive accuracy than the classical linear G kernel. Similarly, Yadav et al. (2024) found that GK and AK surpassed traditional GBLUP models using VanRaden’s linear relationship matrix for two of three traits analyzed in sugarcane. In livestock, Nishio et al. (2023) demonstrated that both GK and AK outperformed GBLUP across all traits in Japanese Black cattle. Despite these encouraging findings, the use of ML kernels in forest tree GP remains limited: GK has been evaluated in only a few cases, while AK has yet to be tested in forest trees. Moreover, although these non-linear models can enhance predictive efficiency, they have not yet been explored within single-step (ssGBLUP) frameworks (e.g. Misztal et al. 2009) in quantitative genetic studies of forest tree species. In this study, we investigate their potential for forest tree improvement, where large and complex genomes and highly polygenic trait architectures present unique challenges for GP applications.

White spruce (*Picea glauca* (Moench) Voss), is a key conifer species in boreal and temperate forests of North America and holds significant commercial value as one of

the primary tree species in Alberta, Canada (Rweyongeza 2011). Tree improvement programs for white spruce increasingly target a broad range of traits, including growth (e.g., height, diameter), wood quality and chemical defense traits (e.g., wood density, secondary metabolites), and climate adaptability (e.g., stable carbon isotope ratio, stomatal conductance) (Cappa et al. 2022b; Galeano and Thomas 2023; Depardieu et al. 2024). Given the complexity of multi-trait tree improvement objectives in a changing climate, evaluating advanced kernel-based GP methodologies such as GK and AK in white spruce represents a critical step toward enhancing predictive performance by incorporating flexible nonlinear representation of underlying genomic factors suited to complex traits. Our study provides the first assessment of these approaches in white spruce, aiming to deliver accurate and cost-effective GP tools to accelerate breeding cycles and support climate-resilient forestry.

In this study, we evaluated the potential of ML-enhanced single- and multi-trait ssGBLUP to improve genomic prediction accuracy for a broad set of 30 complex traits in white spruce, spanning key characteristics in productivity, chemical defense, and climate adaptability, critical to the long-term sustainability of forestry and tree improvement. We compared the performance of the standard single- and multi-trait ssGBLUP model based on VanRaden's linear genomic relationship matrix, used as a benchmark, against two non-linear kernel extensions: the Gaussian kernel (GK) with trait-specific bandwidth kernel average optimization, and the arc-cosine kernel (AK) with increasing depth (1 to 20 layers).

Our specific objectives were to: (i) compare the predictive ability of linear versus non-linear single- and multi-trait ssGBLUP models across a diverse array of traits; (ii) quantify the relative gains in prediction accuracy for traits exhibiting evidence of complex genetic architecture including potential non-additive effects; and (iii) identify the types of traits and kernel configurations that benefit most from integrating ML approaches into the ssGBLUP framework in forest trees. To better understand the ability of non-linear ML kernels to capture non-additive genetic variation, and its relationship to predictive performance, we also fitted an extended single-trait GBLUP model for each of the 30 traits, explicitly incorporating additive, dominance, and epistatic interaction effects (additive $\times$ additive and additive $\times$ dominance).

Materials and methods

Genetic material and trial description

This study was conducted using an open-pollinated (OP) white spruce (*Picea glauca* (Moench) Voss) progeny test established near Calling Lake, Alberta, Canada (55°16'

N, 113°09' W). The trial is part of the Region D1 forest genetic improvement program managed by Alberta Forestry and Parks (previously Alberta Agriculture and Forestry) (FGRMS 2016). The experimental layout followed a randomized complete block design with six replications (blocks), each consisting of 5-tree row plots planted at 2.5 $\times$ 2.5 m spacing. A single border row of trees was planted around the trial which was fenced to protect against herbivory. The breeding population consisted of 150 open-pollinated (OP) families selected from 10 provenances. Further details regarding the test site characteristics and trial design are provided in (Cappa et al. 2022b).

Phenotypic data

This study used a dataset comprising measurements for 30 traits related to productivity, chemical defense, and climate adaptability. A subset of these traits was previously analyzed by Cappa et al. (2022b), and all traits were included in the more recent study by Cappa et al. (2025).

Productivity traits included diameter at breast height (DBH, measured at approximately 1.3 m) and tree height (HT), both assessed at age 30. Wood density (WD) was measured from 5 mm increment cores taken at DBH, with whole-tree WD calculated as a weighted average of ring-level densities, considering ring width and oven-dry weight. Microfibril angle (MFA) was estimated by X-ray diffraction on the radial face of individual growth rings following (Ukrainetz et al. 2008). Two dendrochronological indices, drought resistance (Resistance) and mean drought sensitivity (Sensitivity), were also derived from ring-width series (see Cappa et al. 2025; Sebastian-Azcona et al. 2025). Stable carbon isotope ratio (C13) was measured from ground wood core samples using isotope ratio mass spectrometry at the Alberta InnoTech Stable Isotope Laboratory (in Victoria, British Columbia, Canada). Defense-related traits included seven monoterpenes (α -pinene, β -pinene, camphene, myrcene, limonene, terpinolene, and camphor), and their total concentration (sum of hexane-extractable compounds), as well as 12 polyphenolic compounds (gallic acid, gallic acid, gallic acid, catechin, pungenol, caffeic acid, vanillin, taxifolin, quercetin, naringenin, kaempferol, apigenin, and isorhamnetin), all extracted and quantified from individual-tree phloem tissue. Gas exchange traits included stomatal conductance (gs), intrinsic water use efficiency (WUE), and intercellular CO₂ concentration (Ci), used as indicators of drought adaptability. To correct for environmental variation, a two-step adjustment was applied: (i) the effect of vapor pressure deficit (VPD) was modeled and corrected, using the natural log of VPD when appropriate; and (ii) the effect of hour after sunrise (HAS) was accounted for by regressing HAS on VPD (or log[VPD])

and using the residuals as covariates. For full details, see Wei et al. (2022). Additional methodological details on trait assessment procedures can be found in Cappa et al. (2022b, 2025).

Summary statistics and heritability estimate for each trait are presented in Table 1. A total of 3,935 and 3,936 trees were evaluated for the productivity traits DBH and HT, respectively. For the remaining traits, the number of phenotyped trees varied between 316 and 610, depending on the trait (Supplementary File S1).

Following Cappa et al. (2022b, 2025) logarithmic transformations were applied to MFA, as well as to all monoterpene and polyphenolic compounds (except for gallic acid, catechin, and vanillin), to improve data normality and ensure consistency with previous analyses conducted on the same dataset. Due to spatial heterogeneity within the

trial, and to improve computational tractability, the statistical analyses were conducted using a two-stage approach. In the first stage, phenotypic data were spatially adjusted using the design effects within a pedigree-based *a priori* design model for each trait. Specifically, the adjusted phenotypic data for each trait were obtained by subtracting the estimated replication effects from the original observations for each tree. These design-adjusted phenotypes were subsequently used as response variables in all genomic prediction models evaluated in the second stage (see Statistical analysis below). In this implementation, the uncertainty associated with the adjusted phenotypes was not propagated, and all observations were treated as equally weighted, ignoring differences in precision among individuals and potentially reducing efficiency relative to a single-stage analysis.

Table 1 Summary statistics and heritability estimates (h^2 , and approximate standard errors) for the 30 traits assessed in the white spruce population based on raw phenotypes. Monoterpene concentrations are reported on a dry weight basis. Estimated narrow-sense heritability were obtained using ssGBLUP with standard linear kernel (VanRaden's genomic relationship G matrix)

Type of trait	Trait full name	Trait	Unit	h^2	Mean	SD	Min	Max
Productivity	Tree height	HT	cm	0.44 (0.08)	838.21	196.02	150	1290
	Diameter at breast height	DBH	cm	0.26 (0.10)	12.82	3.53	1	25
	Wood density	WD	kg.m ⁻³	0.42 (0.14)	379.60	29.21	304.07	497.64
	Microfibril angle	MFA^a	°	0.35 (0.16)	21.27	3.97	17.15	50.32
Climate-adaptability	Drought resistance	Resistance	-	0.25 (0.18)	0.57	0.14	0.23	1.20
	Mean drought sensitivity	Sensitivity	-	0.33 (0.17)	0.21	0.06	0.03	0.38
	Average stable carbon isotope ratio	C13	-	0.58 (0.14)	-26.28	0.57	-28.14	-24.37
Defense chemical compounds	α-pinene	α-pinene^a	ng mg ⁻¹	0.50 (0.14)	221.32	165.14	13.99	1029.56
	β-pinene	β-pinene^a	ng mg ⁻¹	0.33 (0.19)	35.81	20.91	9.24	153.40
	camphene	camphene^a	ng mg ⁻¹	0.86 (0.14)	551.15	421.16	30.95	2585.76
	camphor	camphor^a	ng mg ⁻¹	0.29 (0.17)	629.99	587.26	30.23	3842.54
	myrcene	myrcene^a	ng mg ⁻¹	0.76 (0.14)	452.09	466.50	13.79	5644.61
	limonene	limonene^a	ng mg ⁻¹	0.49 (0.15)	606.88	509.85	13.42	3406.85
	terpinolene	terpinolene^a	ng mg ⁻¹	0.60 (0.18)	33.61	20.86	8.17	145.48
	total monoterpenes	total monoterpenes^a	ng mg ⁻¹	0.48 (0.15)	4037.35	2583.12	33.44	15507.53
	gallic acid	gallic acid	mg/kg	0.23 (0.22)	0.80	0.15	0.23	1.55
	galocatechi	galocatechin^a	mg/kg	0.35 (0.20)	154.34	100.09	14.22	487.61
	catechin	catechin	mg/kg	0.25 (0.22)	3332.53	1242.64	721.07	7605.46
	pungenol	pungenol^a	mg/kg	0.22 (0.23)	39.59	19.40	10.63	145.57
	caffeic acid	caffeic acid^a	mg/kg	0.81 (0.17)	6.99	5.33	0.57	39.23
	vanillin	vanillin	mg/kg	0.51 (0.20)	7.42	2.01	2.88	13.46
	taxifolin	taxifolin^a	mg/kg	0.50 (0.23)	20.93	11.98	7.41	80.90
	quercetin	quercetin^a	mg/kg	0.10 (0.02)	16.67	11.28	3.39	70.60
	naringenin	naringenin^a	mg/kg	0.56 (0.18)	2.17	4.47	0.48	38.54
	kaempferol	kaempferol^a	mg/kg	0.18 (0.02)	122.37	90.31	11.20	507.22
	apigenin	apigenin^a	mg/kg	0.52 (0.17)	0.50	0.27	0.14	3.13
	isorhamnetin	isorhamnetin^a	mg/kg	0.19 (0.23)	1.13	0.87	0.20	6.08
Climate-adaptability	Stomatal conductance	gs	-	0.00 (0.02)	0.17	0.10	0.02	0.58
	Intrinsic water use efficiency	WUE	-	0.26 (0.25)	0.08	0.03	0.03	0.19
	Intercellular CO₂ concentration	Ci	-	0.26 (0.25)	242.50	41.29	66.61	322.25

NOTE: ^aLogarithmic transformed for heritability calculation, mean, standard deviation (SD), phenotypic coefficient of variation (CV), minimum (Min.), and maximum (Max.) values.

Genomic data

Genotypes from a genotyping-by-sequencing (GBS) platform (Chen et al. 2013) as described by Cappa et al. (2022b, 2025), were available for 612 trees. After quality filtering for SNPs with $\leq 30\%$ missing data, a minor allele count ≥ 1 , and a maximum site read depth ≤ 70 , a final set of 211,061 SNPs was obtained for the 612 trees. Missing data were imputed using the mean allele frequency (Ratcliffe et al. 2015). Although 612 trees were genotyped in total, the number of genotyped trees with available phenotypic data varied by trait, ranging from 316 to 612. Similarly, the number of genotyped trees included in the validation set also varied by trait, from 136 to 341 (Supplementary File S1).

Statistical analysis

For each trait separately, we fitted the following single-trait ssGBLUP model which simultaneously included both genotyped and non-genotyped trees:

$$y = X\beta + Za + e \quad (1)$$

where y is the $(n \times 1)$ vector of adjusted phenotypes for each individual tree; β is the $(p=10 \times 1)$ vector of fixed systematic effects corresponding to genetic groups defined by provenance; a is the $(q \times 1)$ vector of random additive genetic effects (breeding values) of all trees contributing data, including their ancestors without data; and e is the $(n \times 1)$ vector of random residuals. The incidence matrices X ($n \times p$) and Z ($n \times q$) relate observations to the fixed genetic group effects in β , and the random additive genetic effects for each tree in a . Residuals are assumed to follow a normal distribution: $e \sim N(0, I\sigma_e^2)$, where I ($n \times n$) is an identity matrix and σ_e^2 is the residual variance. Additive genetic effects are assumed to follow a multivariate normal distribution: $N(0, H\sigma_a^2)$, where σ_a^2 is the variance of additive genetic effects, and H is the combined pedigree- and marker-based relationship matrix of dimension $q \times q$ (Legarra et al. 2009), defined as:

$$H = \begin{bmatrix} A_{11} + A_{12}A_{22}^{-1}(K - A_{22})A_{22}^{-1}A_{21} & A_{12}A_{22}^{-1}K \\ KA_{22}^{-1}A_{21} & K \end{bmatrix}$$

where A_{11} , A_{12} , A_{21} , and A_{22} are sub-matrices of the pedigree-based relationship matrix A (Henderson 1984). Subscripts 1 and 2 denote non-genotyped and genotyped individuals, respectively. Following (Momen et al. 2022), the matrix K can be any valid (positive semi-definite) kernel matrix of order equal to the number of genotyped individuals, constructed from marker information to represent

the covariance structure among them. We evaluated one linear (G) and two non-linear kernels (GK and AK) in this study, as detailed in the following subsections of the Materials and Methods. To scale differences between genomic and pedigree-based information, the scalar parameter λ was set to 0.95 for all K matrices (Aguilar et al. 2010), ensuring a consistent weighting across traits and models, and avoiding the need for trait-specific tuning of this parameter.

To illustrate and compare the potential of multi-trait ssGBLUP prediction using linear and non-linear kernels, and evaluate model performance in this empirical context, we selected a representative subset of four key traits: one related to productivity (height, HT), one defense-related chemical compound (camphene), and two climate-adaptability traits (resistance and water-use efficiency, WUE). The multi-trait ssGBLUP model was specified as:

$$\begin{bmatrix} y_1 \\ \vdots \\ y_4 \end{bmatrix} = \begin{bmatrix} X_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & X_4 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_4 \end{bmatrix} + \begin{bmatrix} Z_{a_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & Z_{a_4} \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_4 \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_4 \end{bmatrix} \quad (2)$$

where, $y_1, \dots, y_4$ are the vectors of individual tree adjusted phenotypes for traits 1 to 4 (i.e., 1=HT, 2=Resistance, 3=camphene, and 4=WUE); $\beta' = [\beta'_1, \dots, \beta'_4]$ is the vector of fixed effects corresponding to genetic groups defined by provenance; and $a' = [a'_1, \dots, a'_4]$ is the vector of random additive genetic effects (i.e., breeding values). The additive effects are assumed to follow a multivariate normal distribution: $a \sim N(0, \sum_a \otimes H)$, where $\sum_a$ is the 4×4 (co)variance matrix among traits, and H is the combined pedigree- and marker-based relationship matrix defined above. The residual vector is defined as $e' = [e'_1, \dots, e'_4]$ with $e \sim N(0, R_0 \otimes I)$, where R_0 is the 4×4 residual (co)variance matrix and I is an identity matrix. Both $\sum_a$ and R_0 were assumed to have unstructured (co)variance forms. The design matrices X_1 to X_4 , and Z_{a_1} to Z_{a_4} relate the adjusted phenotype to the means of the genetic group effects in β , and the additive genetic effects in a .

Additionally, to investigate the contribution of different sources of genetic variance, we fitted a multi-component GBLUP model separately for each of the 30 traits, using only genotyped trees. The model included additive, dominance, and epistatic interaction effects, specifically, additive $\times$ additive and additive $\times$ dominance interactions, and was specified as:

$$y = X\beta + Z_1a + Z_2d + Z_3aa + Z_4ad + e \quad (3)$$

where vectors y and β were previously defined in model [1], where y is the vector of adjusted phenotypes for each

genotyped individual tree, and β is a vector of fixed effects for provenance-based genetic groups. The random additive genetic effects a for genotyped trees were assumed to follow a multivariate normal distribution $a \sim N(0, G\sigma_a^2)$, where G is the additive genomic relationship matrix computed using VanRaden's method (see next subsection below). Dominance effects d were assumed to follow $d \sim N(0, D\sigma_d^2)$, with D being the dominance relationship matrix calculated using the Gmatrix() function from the "AGHmatrix" R package (Amadeu et al. 2016), following Vitezica et al. (2013). The epistatic interaction effects, aa and ad representing additive $\times$ additive and additive $\times$ dominance interactions, were modeled as $aa \sim N(0, E_{aa}\sigma_{aa}^2)$, and $ad \sim N(0, E_{ad}\sigma_{ad}^2)$, respectively. The matrices E_{aa} and E_{ad} were constructed using the Hadamard (element-wise) product of the additive and dominance genomic relationship matrices, as described by Muñoz et al. (2014). The residuals were assumed to be normally distributed as $e \sim N(0, I\sigma_e^2)$. Variance components were estimated using restricted maximum likelihood (REML) via the Expectation-Maximization (EM) algorithm implemented in the remlf90() function from the breedR package in R (Muñoz and Sanchez 2025), followed by one iteration of the Average Information (AI) algorithm to compute the approximate standard errors of variance components (Cha-teigner et al. 2020). The design matrices Z_1 , Z_2 , Z_3 , and Z_4 , in model [3] are incidence matrices linking the vector of adjusted phenotypes y to the random effects: a , d , aa , and ad respectively. For each trait, we computed the proportion of phenotypic variance explained by the additive, dominance, and epistatic (additive $\times$ additive and additive $\times$ dominance) components, as well as the residual variance, to evaluate the relative contribution of each source of variation to the total genetic architecture.

Linear kernel

The linear kernel applied in this study corresponds to the widely used genomic relationship matrix (G) in ssGBLUP, computed following the method proposed by VanRaden (2008). The VanRaden's genomic relationship G matrix (G) is defined as:

$$G = \frac{WW'}{2 \sum_k p_k(1 - p_k)}$$

where, W is the centered matrix of SNP covariates (i.e., marker genotypes coded as 0, 1, or 2), and p_k is the current (or observed) allele frequency of the genotyped trees for marker k . The resulting G matrix was used to construct the combined relationship H matrix in the ssGBLUP framework,

serving as the baseline (benchmark) model for comparison with non-linear kernel approaches. The G matrix was computed in R (<http://www.rproject.org>) using the Gmatrix() function from the "AGHmatrix" package (Amadeu et al. 2016). This G matrix was then used to construct the combined relationship matrix (H) for the ssGBLUP approach, using the H.inverse() function from the "ASRgenomics" package (Gezan et al. 2022).

Machine learning kernels

We employed two machine learning kernels: the Gaussian kernel (GK) (Gianola et al. 2006; Gianola and Van Kaam 2008) and the arc-cosine kernel (AK) (Cuevas et al. 2019). The Gaussian kernel is defined as:

$$GK_{ij} = \exp(-h \|w_i - w_j\|^2)$$

where $\|w_i - w_j\|^2 = \sum_{k=1}^m (w_{ik} - w_{jk})^2$ is the squared-Euclidean distance between the SNP marker profiles of trees i and j , and h is a positive scalar (bandwidth parameter) that controls the rate at which similarity decays with genetic distance. Following Alves et al. (2019) (see Supplementary Method 2), the squared Euclidean distances were derived directly from the additive relationship G matrix, allowing efficient computation of marker-based distances.

The choice of the bandwidth parameter h is critical, as it influences the smoothness and locality of the kernel (de los Campos et al. 2010): smaller values of h yield broader kernels with higher covariances across individuals, while larger values result in more localized kernels that emphasize similarity among closely related individuals. Increasing the bandwidth parameter produces a sharper decay in covariance with decreasing additive relationship, which may better capture non-additive and epistatic interaction effects (Supplementary File S2) (Alves et al. 2019). To address the uncertainty in selecting a single optimal h , we implemented a kernel averaging strategy (also referred to as multiple kernel learning) as proposed by de los Campos et al. (2010) and further applied by Momen et al. (2022) in the ssGBLUP context. Specifically, six Gaussian kernels were computed using different h values: $\{0.1, 0.25, 0.5, 1, 1.5, 2\}$, selected to span a range of bandwidths, capturing different scales of genetic similarity. Each yielding in a kernel (GK_1 to GK_6) with distinct properties. These six kernels were then combined into a single composite kernel by weighting each one according to its estimated variance component. The final GK matrix was the weighted sum of these individual kernels, where the weights reflected the proportion of total genetic variance explained by each kernel (see details in Momen et al. 2022). A custom R script was used to compute the six GK

matrices and the final averaged GK matrix. This composite GK matrix was then used to construct the $\mathbf{H}$ matrix in the ssGBLUP framework, using the `H.inverse()` function from the “ASRgenomics” package (Gezan et al. 2022).

The arc-cosine kernel (AK) was computed following the approach described by Cuevas et al. (2019), who proposed it as an alternative to GK for capturing complex and hierarchical interactions in genomic data. In this method, the similarity between two individuals i and j is defined by the angle θ between their corresponding genotype vectors w_i and w_j , as introduced by Cho and Saul (2009):

$$\theta_{ij} = \cos^{-1} \left(\frac{w_i \bullet w_j}{\|w_i\| \|w_j\|} \right)$$

where $\bullet$ denotes the inner product and $\|w_i\|$ is the norm of the vector w_i . For a single layer ($l=1$), the AK between individuals i and j is given by Cuevas et al. (2019):

$$AK_{ij}^{l=1}(w_i, w_j) = \frac{1}{\pi} (\|w_i\| \|w_j\| J(\theta_{ij}))$$

where π is the mathematical constant (3.1418...), and $J(\theta) = [\sin(\theta) + (\pi - \theta) \cos(\theta)]$.

Because the AK can be interpreted as the inner product in an induced feature space $\Phi(w_i) \bullet \Phi(w_j)$, Cho and Saul (2009) showed that it can mimic computations in deep multilayer threshold networks through a recursive formulation, where the kernel in one layer is a function of the kernel in the previous layer. The kernel at layer $l+1$ is given by:

$$AK_{ij}^{l+1}(w_i, w_j) = \frac{1}{\pi} \left[AK^{(l)}(w_i, w_i) AK^{(l)}(w_j, w_j) \right]^{1/2} \left(J(\theta_{ij}^{(l)}) \right)$$

where

$$\theta_{ij}^{(l)} = \cos^{-1} \left\{ AK^{(l)}(w_i, w_j) \left[AK^{(l)}(w_i, w_i) AK^{(l)}(w_j, w_j) \right]^{-1/2} \right\}.$$

The base AK with layer=1 (AK^1), as well as the recursive computation of higher-order layers up to $l=20$ (i.e., AK^2 to AK^{20}) were implemented for each trait using the R script provided in Cuevas et al. (2019).

As highlighted by Crossa et al. (2019), it is important to note that the diagonal elements of the AK matrix are not uniform, reflecting heterogeneous variances in the additive genetic values. In contrast, the GK matrix has a constant diagonal, indicating homogeneous variances. This distinction may offer a theoretical advantage for AK in capturing the complexity of inter-individual genetic relationships.

For the multi-trait analysis, we also implemented the kernel averaging approach by de los Campos et al. (2010). Specifically, the final GK matrix was constructed as a weighted sum of the individual GK matrices obtained from the single-trait models for the four representative traits (HT, Resistance, camphene, and WUE), and this composite matrix was then used to build the $\mathbf{H}$ matrix for the multi-trait ssGBLUP model. Similarly, the final AK matrix used in the multi-trait analysis was derived as the weighted sum of the trait-specific AK matrices corresponding to the best-performing depth (i.e., number of layers) in the single-trait analysis: 4 for HT, 8 for Resistance, 15 for camphene, and 20 for WUE (see below).

Cross-validation analysis

A ten-fold cross-validation was performed across all 30 traits. In each fold, one randomly selected subset of trees served as the validation set, while the remaining nine formed the training set. Every tree with phenotypic data appeared in the training set at least once across the ten folds. Because not all 30 traits were measured in every tree (Supplementary File S1), the validation set was restricted to trees with genotype data and no more than four missing phenotypes (i.e., trees with at least 26 observed traits), ensuring sufficient phenotypic information for reliable multi-trait prediction while maintaining an adequate sample size. This resulted in a final validation set of 341 trees. Each fold included approximately one-tenth of these trees. Importantly, the same validation set was used across all traits, meaning the same phenotypic values were masked in each fold. For the multi-trait analysis, the same trees were masked as in the single-trait cross-validation, ensuring consistency across analyses.

Given that the “true” breeding value is unknown, reference breeding values were estimated using pedigree information and all available trees, and were treated as proxies for the “true” breeding values (Cappa et al. 2019). Then, the prediction ability (PA) was calculated as the average Pearson correlation between these reference values, and the predicted values for trees in the validation set for each kernel ssGBLUP model. Therefore, PA should be interpreted with caution, as the reference values are themselves subject to estimation error. Standard errors of PA were thus provided across cross-validation folds to quantify uncertainty and facilitate robust interpretation of PA.

For each trait, three single-trait ssGBLUP models were fitted using the combined $\mathbf{H}$ matrix constructed from either the linear genomic relationship matrix (G) or one of the two machine learning-based kernels (GK and AK) described above. The same three kernels were also applied

in the multi-trait ssGBLUP model to evaluate predictive performance in a multi-trait context. Model fitting was performed using the `remlf90()` function from the “breedR” package (Muñoz and Sanchez 2025). Estimation followed the Expectation-Maximization (EM) algorithm, with one final iteration of the Average Information (AI) algorithm to obtain approximate standard errors of the variance components (Chateigner et al. 2020). Cross-validation analyses were conducted and automated using a customized R script tailored to each trait.

Results

Partition of the genetic variance components across trait

To better understand the genetic architecture underlying each trait, and to relate these patterns to the differential predictive ability observed across linear and non-linear models, we decomposed the total phenotypic variance using a GBLUP model that incorporated additive, dominance, and epistatic interaction (additive $\times$ additive and additive $\times$ dominance) effects, based on a standard linear kernel (VanRaden’s genomic relationship matrix, **G**). As shown in Fig. 1 (and Supplementary File S3), additive genetic variance (blue) tended to be the predominant component for the majority of traits (21 out of 30), often explaining over 40% of the phenotypic variance. However, non-additive effects also played a substantial role in several traits. In particular, dominance variance (orange) accounted for more than ~50% of the total variance in traits such as resistance, β -pinene, pungenol, quercetin, and kaempferol. Additionally, all physiological traits (gs, WUE, and Ci), along with three chemical defense traits (limonene, total monoterpenes, and gallic acid), showed relatively high levels of epistatic variance (green), each accounting for more than ~25% of the total phenotypic variance. Traits in Fig. 1 are ordered by decreasing epistatic contribution, emphasizing the diversity in genetic complexity across traits.

Predictive ability across kernels and traits using single-trait ssGBLUP

The predictive performance of three single-trait ssGBLUP models, based on the linear additive genomic relationship matrix (**G**), the Gaussian kernel (**GK**), and the deep arccosine kernel (**AK**), was evaluated across 30 traits related to productivity, defense, and climate adaptability (Fig. 2; Supplementary File S4). Predictive ability varied by trait and kernel type. Overall, the **AK** model outperformed both **G** and **GK** for most traits, particularly those related to defense

and climate response, including physiological traits. Specifically, **AK** yielded the highest predictive ability in 29 out of 30 traits. The most pronounced gains were observed for gs (from 0.275 to 0.349, 26.9%), Ci (from 0.413 to 0.511, 23.5%), WUE (from 0.393 to 0.483, 23.1%), and limonene (from 0.407 to 0.450, 10.7%), all of which are associated with physiological processes or secondary metabolites (Figs. 2 and 3). Moderate improvements (5.8–9.4%), corresponding to predictive differences of 0.020–0.051, were found for the remaining monoterpenes, three polyphenols (vanillin, galocatechin, and pungenol), as well as for the productivity trait WD (from 0.513 to 0.542, 5.6%). The remaining polyphenols and productivity traits (HT and MFA) showed more modest gains, with predictive ability increasing by less than 5.6% (predictive differences < 0.029). In contrast, for DBH, the linear **G** kernel outperformed both non-linear models, with predictive ability decreasing from 0.756 to 0.729 and 0.745 for **GK** and **AK**, respectively.

The **GK** showed more moderate gains (predictive differences < 0.028; < 6.7%), exceeding the linear **G** kernel in 20 traits, with consistently smaller effect sizes than **AK**. For example, Ci showed a 6.7% gain with **GK** (from 0.413 to 0.441), whereas kampferol and catechin showed slight reductions compared to **G** (–3.3 and –2.8%, respectively). On average, **GK** had a narrower impact on predictive ability than **AK** (Figs. 2 and 3). However, the increases in PA provided by **GK**, relative to the linear **G** kernel, were moderately correlated with those of **AK** across the 30 traits (Pearson correlation $r=0.59$), suggesting that both kernels may capture partially overlapping signal. An exception was observed for gs, where **AK** exhibited relatively larger gains (predictive difference of 0.074), whereas **GK** showed no improvement.

For the productivity growth-related traits (HT and DBH) and wood properties (WD and MFA), predictive abilities were, in general, relatively high across all kernels, but improvements from non-linear kernels were limited. For instance, HT showed similar predictive ability across models ($G=0.707$; $GK=0.705$; $AK=0.717$). The same pattern was observed for MFA, where all models performed similarly and showed no gain in predictive accuracy with deeper kernels.

Predictive ability across kernels using multi-trait ssGBLUP

The performance of multi-trait ssGBLUP models fitted with **G**, **GK** and **AK** was evaluated using a representative subset of four traits: HT, Resistance, camphene, and WUE. As shown in Fig. 4, for HT, the linear kernel (**G**) yielded higher predictive ability (mean = 0.630) than both **GK** (0.575) and **AK** (0.576), as indicated by different letters (p -value < 0.05).

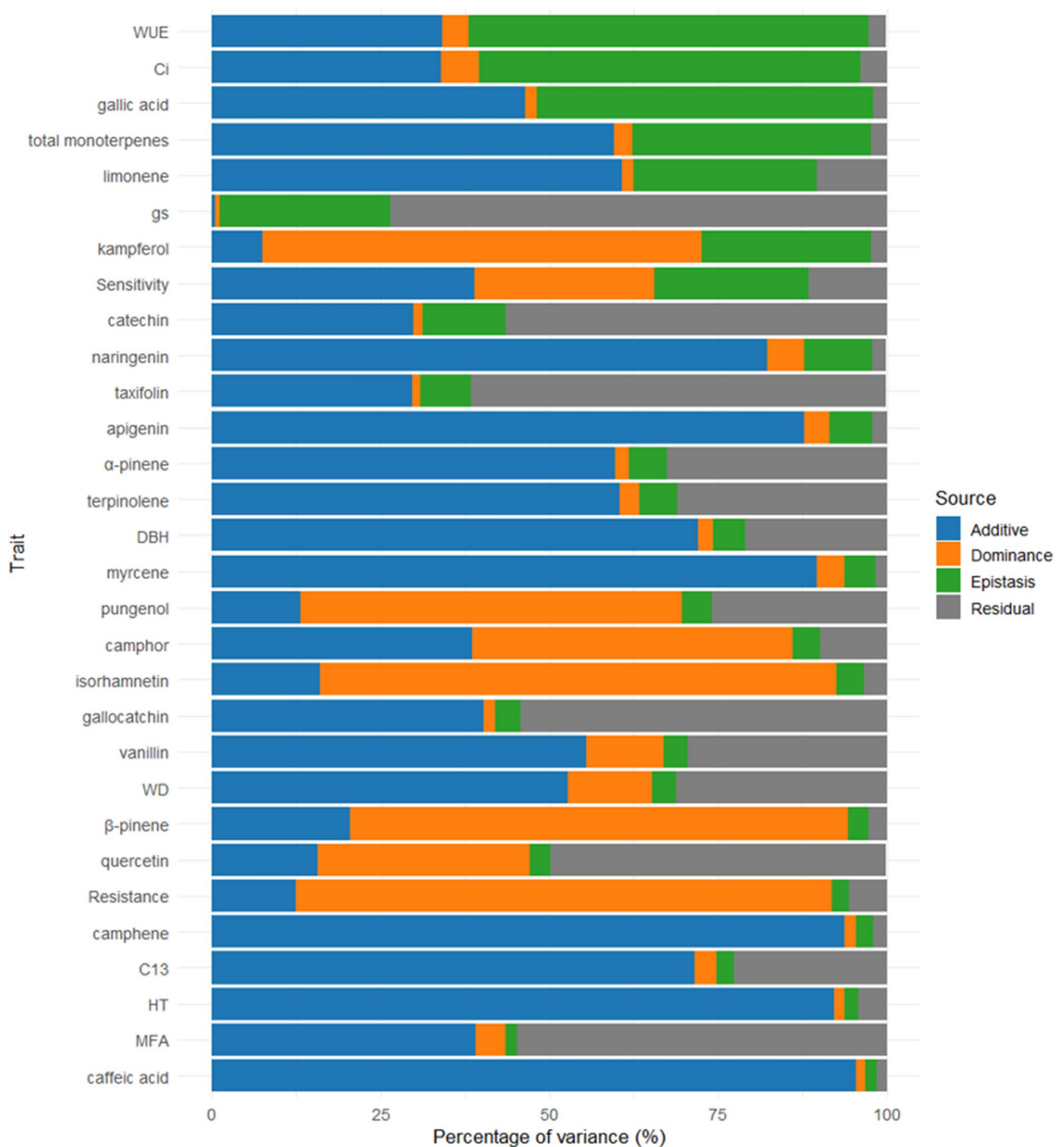


Fig. 1 Partitioning of estimated variance components for each trait based on the GBLUP model using a standard linear kernel (VanRaden's genomic relationship matrix, G). Variance components included additive, dominance, epistatic (additive $\times$ additive and additive $\times$ domi-

nance), and residual effects. Traits are ordered from highest to lowest according to the proportion of total genetic variance attributed to epistasis, including both additive $\times$ additive and additive $\times$ dominance interactions. Trait abbreviations are defined in the main text

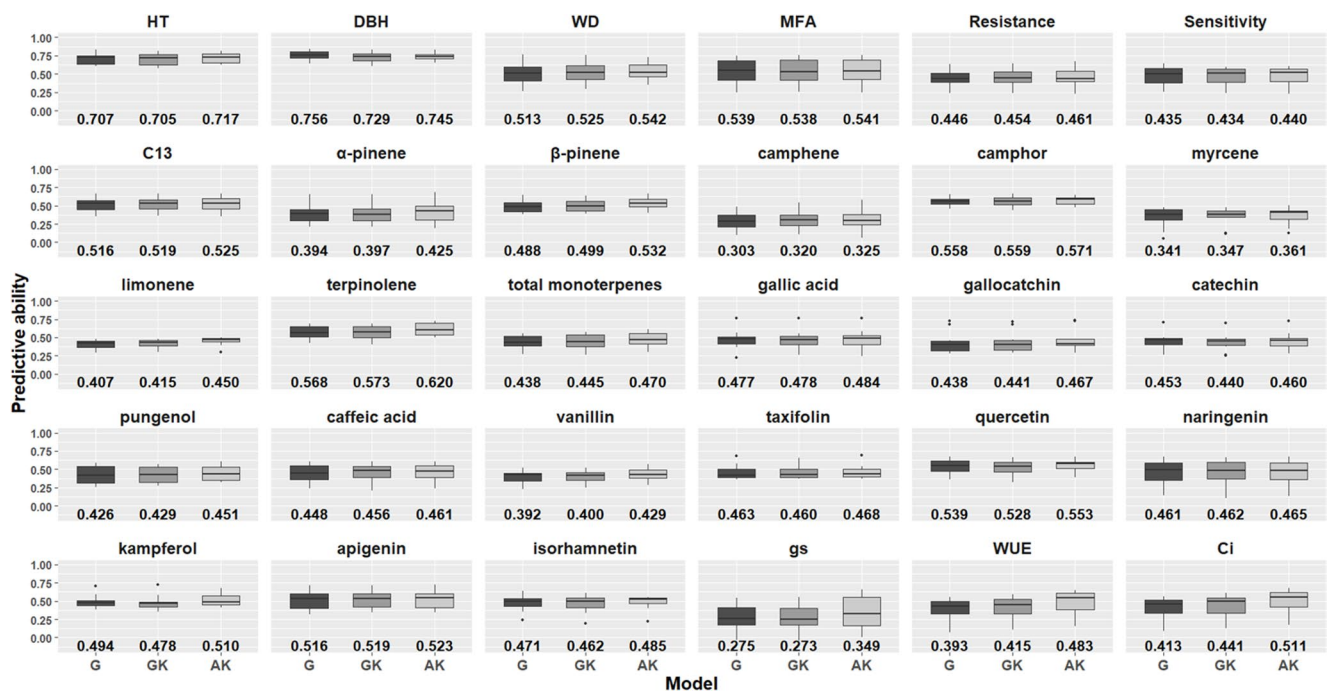


Fig. 2 Average predictive ability for the 30 studied traits using single-step GBLUP models with three genomic relationship kernels: the standard linear kernel (VanRaden's, G), and two non-linear kernels (the Gaussian kernel -GK- and the deep arc-cosine kernel -AK-). For GK method, the bandwidth parameter (h) was estimated for each trait

using a kernel averaging approach. Results for AK method correspond to the highest predictive ability achieved across 1 to 20 layers. Average predictive abilities are shown below each boxplot. Trait abbreviations are defined in the main text

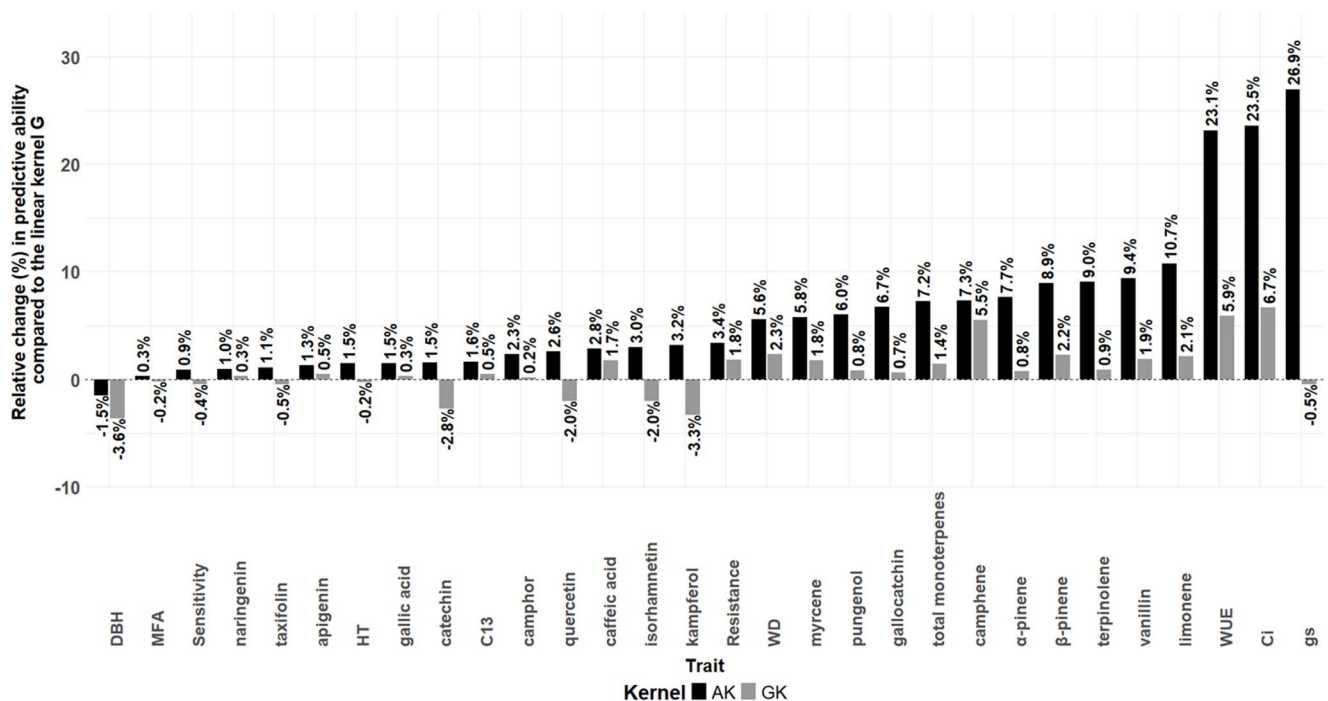
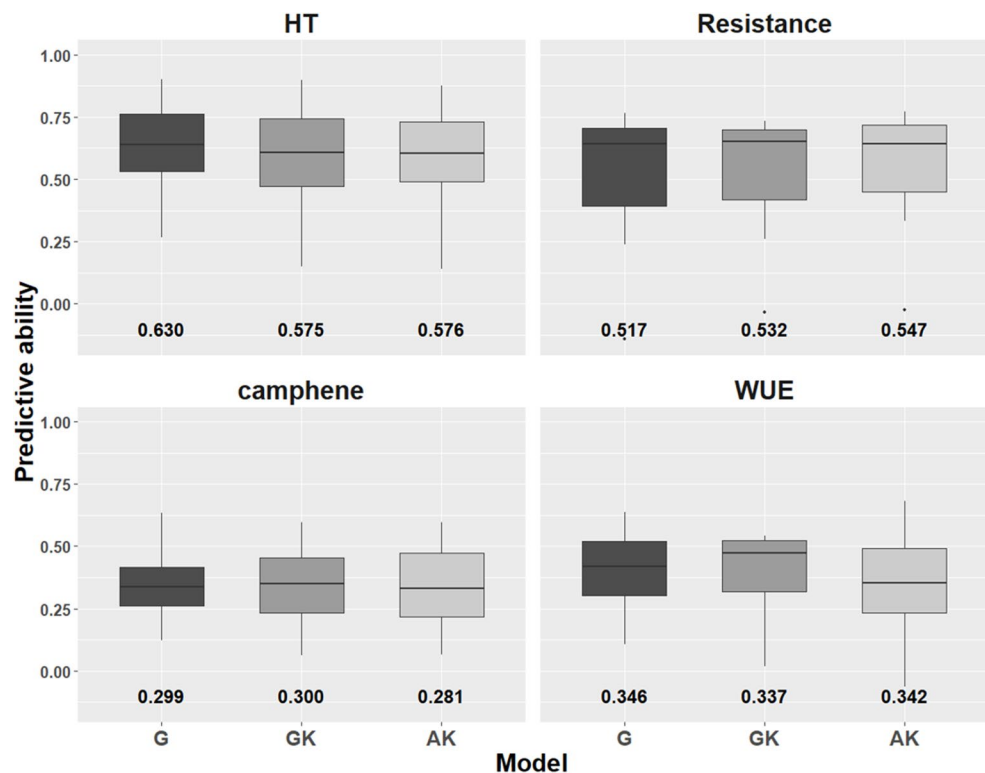


Fig. 3 Relative change in predictive ability (%) for the non-linear Gaussian kernel (GK) and deep arc-cosine kernel (AK) compared to the standard linear kernel (VanRaden's, G). Trait abbreviations are defined in the main text

Fig. 4 Average predictive ability for the four representative studied traits using multi-trait single-step GBLUP models with three genomic relationship kernels: the standard linear kernel (VanRaden's, G), and two non-linear kernels (the Gaussian kernel -GK- and the deep arc-cosine kernel -AK-). Average predictive abilities are shown below each boxplot. Trait abbreviations are defined in the main text



In contrast, for the other three traits: Resistance, camphene, and WUE, predictive abilities were similar across kernels, with values ranging from 0.281 to 0.547. Interestingly, for Resistance, AK slightly outperformed both G and GK (mean=0.547 vs. 0.517 and 0.532, respectively). In the case of camphene, GK achieved the highest predictive ability (0.300), closely followed by G and AK. For WUE, all kernels performed similarly, with average predictive abilities around 0.340. These results suggest that while the linear kernel may perform best for certain traits such as HT, non-linear kernels can yield comparable prediction accuracy for other traits in a multi-trait context, particularly those with potentially complex genetic architectures.

When comparing the results of multi- *versus* single-trait ssGBLUP models for the four representative traits (HT, Resistance, camphene, and WUE), we observed consistent predictive ability for HT, camphene, and WUE, and slightly improved predictive ability for Resistance under the multi-trait framework (Fig. 2 vs. Figure 4). For HT, predictive ability remained stable across kernels, with the linear kernel (G) performing best in both approaches (0.630 in multi-trait vs. 0.707 in single-trait). In contrast, for Resistance, multi-trait models improved prediction accuracy from 15.9 to 18.7%, with the AK achieving notably higher accuracy (0.547) compared to single-trait models (0.461). These results suggest that leveraging information from genetically correlated traits enhances prediction performance for traits with lower heritability ($h^2=0.25$, Table 1). For camphene,

predictive abilities were overall similar across single- and multi-trait models, with minor increases in multi-trait models (e.g., AK: 0.342 in multi-trait vs. 0.325 in single-trait).

Discussion

Maximizing the predictive performance of statistical models remains a central goal in genomic selection, and this performance is shaped by various factors, including traits' underlying genetic architecture. While additive models have been found to account for most of the genetic variance in various datasets (Hill et al. 2008), recent GBLUP studies in both animal breeding (Su et al. 2012; Nishio and Satoh 2014; Momen et al. 2018) and plant breeding (Lyra et al. 2017; Wang et al. 2017; Yadav et al. 2021) have demonstrated that using either parametric models with distinct kernels to explicitly capture additive, dominance, and epistatic effects, or machine learning-based alternatives such as the Gaussian kernel (GK) and the arc-cosine kernel (AK), can significantly enhance prediction accuracy compared to models that exclusively account for additive effects alone.

This study evaluated two non-linear machine learning kernels, GK and AK, for their potential to enhance predictive accuracy within the ssGBLUP framework for white spruce. Their predictive performance was compared against the standard linear genomic relationship matrix (G) derived using VanRaden's method, across 30 traits key to

productivity, defense, and climate-adaptability. Our results indicate that the two machine learning kernels (GK and AK) generally achieved higher predictive ability than the linear G kernel for most traits (29 out of 30 traits; see Figs. 2 and 3), although the magnitude of these differences varied across traits. More specifically, the greatest improvements were found in traits that exhibit high levels of epistatic variance and certain chemical defense compounds like limonene, total monoterpenes, and gallic acid (Fig. 3). These findings highlight the importance of modeling non-additive effects, and suggest that strictly additive models based on linear kernels may be insufficient to fully capture the complexity of genetic architectures, thereby limiting predictive performance.

Consistent with previous pedigree- and marker-based studies in forest trees (Araújo et al. 2012; Muñoz et al. 2014; Tan et al. 2018; Calleja-Rodriguez et al. 2021; Garel and Vigneron 2022; Dong et al. 2024), we found that although additive genetic variance remained the predominant component for most traits, non-additive components such as dominance accounted for over 50% of the variance in compounds like pungenol and β -pinene, while epistatic variance can exceed 25% for traits such as stomatal conductance (gs), water-use efficiency (WUE), and intercellular CO₂ concentration (Ci) (Fig. 1 and Supplementary File S3). Several GBLUP studies in forest species have demonstrated that including these non-additive effects, whether explicitly by parametric or using semi-parametric GK, can improve genomic prediction performance (Denis and Bouvet 2013; de Almeida Filho et al. 2016; Pégard et al. 2020; Thavamanikumar et al. 2020; Calleja-Rodriguez et al. 2021; Paludeto et al. 2021; Thumma et al. 2021; Dong et al. 2024). However, results have been trait- and context-specific. For instance, in black poplar, Pégard et al. (2020) found that including dominance effects in additive models did not significantly improve prediction accuracy. Similarly, de Almeida Filho et al. (2016) reported only modest gains in predictive accuracy for tree height when dominance was added. Yet, their simulation studies, based on a genetically similar population, revealed that when dominance accounted for 20% or more of the phenotypic variance, models including both additive and dominance effects outperformed additive-only models.

Although less extensively studied than dominance, the role of epistatic interactions in prediction has shown considerable promise. Calleja-Rodriguez et al. (2021) reported that epistatic variance, including first order epistatic effects (additive $\times$ additive and additive $\times$ dominance and dominance $\times$ dominance), accounted for 2.3 to 34.5% of the phenotypic variation in traits such as growth, wood density, and modulus of elasticity in *Pinus sylvestris*. Including these effects explicitly through parametric models improved the prediction of growth and wood quality traits', though not for

modulus of elasticity. The authors suggested that accounting for epistasis within full-sib families may improve total genetic value estimation for certain traits. Recently, in *Larix kaempferi* (Lam.) Carrière, Dong et al. (2024) showed that additive $\times$ additive epistasis effects explained a substantial proportion of the phenotypic variance (50.8–91.3%) for traits such as growth, competitive ability, and acoustic velocity. In their study, predictive ability was significantly improved when using the non-linear GK compared to standard additive models or those including both additive and dominance effects. The GK applies an exponential transformation to additive genetic effects, which allows it to implicitly capture certain forms of epistatic interactions, even when these are not explicitly modeled (Jiang and Reif 2015; Varona et al. 2018).

In our study, although both non-linear kernels (GK and AK) generally achieved higher predictive ability than the linear kernel G across most traits, the magnitude of the improvement varied depending on the kernel used. AK consistently delivered greater gains than GK (up to 27 vs. 7%, respectively) (Fig. 3), corresponding to larger differences in predictive ability (~ 0.10 for AK vs. ~ 0.03 for GK), especially for traits with pronounced epistatic variance components (Fig. 1). Consistently, the relative gain in predictive ability of AK over G showed a moderate positive correlation with the proportion of epistatic variance across traits (Pearson correlation $r=0.59$). The superior performance of AK can be attributed to its theoretical properties. Unlike GK, which measures similarity based on Euclidean distance and works most effective for smooth, local interactions, AK emphasizes angular similarity and is mathematically equivalent to an infinite-width neural network with rectified polynomial activations (Cho and Saul 2009). This structure enables AK to implicitly capture higher-order non-linear interactions more effectively, as reflected in the much-improved prediction performance with deeper kernels (layer=20, Supplementary File S4) for the traits with significant component of epistatic variance, such as gs, WUE, and Ci (Figs. 1 and 3). These results suggest that the success of machine learning-based genomic prediction depends not only on the presence of non-additive genetic variance but also on the compatibility between the kernel's functional form and the trait's genetic architecture. However, as highlighted by Montesinos-López et al. (2021a, b), the benefits of using kernel methods over traditional linear models are context-dependent, influenced by both the characteristics of the dataset and the choice of kernel function. In the present study, analyses were conducted within a single breeding population using within-population cross-validation; therefore, the observed improvements in predictive ability reflect performance under a specific genetic and environmental context. In this sense, the preceding may or may not hold in

practice, and improvements in prediction accuracy should be interpreted with caution as evidence of non-additive effects. Moreover, their performance under more general deployment scenarios (e.g., across-site prediction or across generations) remains to be evaluated.

Deep learning approaches are a class of ML methods within artificial intelligence (AI) that rely on successive nonlinear transformations of the input data (Montesinos-López O. et al. 2021). In these models, information is processed through multiple layers, where each layer represents increasingly complex features derived from the original inputs (Cuevas et al. 2019). In the context of AK, this layered structure can be interpreted as progressively enriching the representation of relationships among individuals as model depth increases. From a statistical perspective, increasing the number of layers progressively introduces more complex nonlinear feature mappings (Cho and Saul 2009), allowing the model to capture complex non-additive genomic effects. In practice, the number of layers can be selected based on predictive performance or marginal likelihood criteria, with improvements in prediction associated with higher values of the marginal likelihood (Cuevas et al. 2019). In our study, we selected the number of layers based on predictive performance, using the highest predictive ability achieved across models ranging from 1 to 20 layers, with optimal depth varying across traits (Supplementary File S4).

In this study, epistatic effects were implicitly captured through non-linear kernel methods rather than being directly parameterized, as in explicit approaches such as model [3]. In explicit approaches, epistasis is often modeled via constructed relationship matrices (e.g., using Hadamard products of additive and dominance relationship matrices) (Varona et al. 2018), which aim to isolate specific interaction components under assumptions such as orthogonality among genetic effects (Vitezica et al. 2017). These assumptions, particularly the absence of covariance between additive and epistatic effects, are expected to hold only under Hardy-Weinberg equilibrium, conditions that are rarely met in operational breeding populations. Moreover, as noted by Alves et al. (2019), additive $\times$ additive epistatic relationship matrices built from Hadamard products do not clearly separate dominance from epistasis contributions (Martini et al. 2016), further underscoring the limitations of explicitly partitioning genetic effects. In contrast, the non-linear kernel-based methods do not attempt to disentangle individual genetic components. Instead, they capture overall non-linear patterns in the data, which may include epistatic interactions, without requiring the construction of explicit relationship matrix or imposing restrictive assumptions (Varona et al. 2018). Importantly, under non-linear kernel-based formulations, the random genetic effect no longer represents a strictly additive breeding value but rather a

more general latent function defined by the chosen kernel. Consequently, while non-linear kernels may implicitly capture signals consistent with non-additive genetic effects, the resulting predictions do not provide a separable or directly interpretable partition of genetic variance components. Therefore, improvements in predictive performance should be interpreted with caution, as they may reflect increased model flexibility rather than a clear identification of underlying non-additive genetic architecture. Beyond this general limitation, a related issue arises in multi-kernel approaches, as kernels are not mutually orthogonal and therefore capture overlapping sources of variation, which complicates the interpretation of their individual contributions when combined.

Here, we introduce for the first time the ssGBLUP framework with non-linear kernels in forest tree breeding. To date, Momen et al. (2022) represent the only comparable study in animal breeding, showing that kernel choice can notably impact prediction accuracy, especially when over 60% of the population is genotyped. Therefore, differences between linear and non-linear kernels appear to be influenced by the proportion of genotyped individuals in the target population. By comparison, our white spruce dataset exhibited a more heterogeneous population structure across traits. For growth traits (HT and DBH), only ~16% of trees with phenotype and pedigree information were genotyped, whereas for most other traits, genotyped and phenotyped individuals largely overlap (Supplementary File S1). The slightly better performance of the linear kernel for DBH, together with the minimal differences observed for HT, are worth highlighting, given their importance as key productivity traits; however, these differences should be interpreted with caution. In line with Momen et al. (2022), differences in predictive accuracy between linear and non-linear kernels tend to diminish when the proportion of genotyped individuals is low. For these traits, the limited proportion of genotyped individuals substantially restricts the amount of genomic information available to inform the models. Consequently, the differences observed between kernels (-3.6% to 1.5; Fig. 3) likely reflect a reduced capacity to capture non-linear genetic patterns, rather than a true superiority of one approach over another.

In this context, although the condition of a high proportion of genotyped individuals (>60%; Momen et al. 2022) is met for most traits in our study, the absence of additional phenotypic information from non-genotyped individuals reduces the potential advantage of ssGBLUP over standard GBLUP in single-trait analyses, as both approaches rely on essentially the same set of individuals and information. However, in multi-trait analyses, the phenotypic information from non-genotyped but phenotyped individuals for HT in both training and validation populations, contributed to

improved predictive accuracy for traits such as Resistance, camphene, and WUE, which had comparatively smaller training populations. This highlights an additional benefit of the ssGBLUP framework when integrating complementary trait information in multi-trait setting, under limited phenotypic records, consistent with findings in *Eucalyptus grandis* (Jurcic et al. 2023). Taken together, in light of our multi-trait results, it appears that the multi-trait linear model may benefit more strongly from the additional information provided by correlated traits than from the gains associated with non-linear kernels, suggesting that, in this multi-trait setting and for this dataset, the observed improvements in predictive accuracy are primarily driven by the incorporation of cross-trait information rather than by kernel choice.

It is important to note that our results also revealed underperformance of GK in the single-trait setting, and of AK and GK in multi-trait analysis, compared to the corresponding linear kernel. These reductions in accuracy may be partly due to the multi-kernel averaging approach used to construct GK in the single-trait and GK and AK in multi-trait models. A key limitation of this approach lies in how weights are assigned to each kernel, as they are based on variance components estimated from multi-kernel fitting. As noted Momen et al. (2022), “*since the kernels are not mutually orthogonal, the weights placed to the individual kernels may not produce the best possible average kernel*”. This limitation could lead to suboptimal kernel combinations and, consequently, reduced predictive performance. Further investigation is needed to refine kernel weighting strategies and improve the effectiveness of multi-kernel models in this context. One promising alternative to kernel averaging is the approach recently proposed by Melsen et al. (2025), which treats the bandwidth parameter of the non-linear GK as an estimable parameter within a restricted maximum likelihood framework.

Conclusion

Overall, our findings demonstrate that non-linear kernels can improve genomic prediction accuracy across a wide range of traits in a Canadian white spruce breeding program, with clear implications for other tree improvement efforts. However, these findings are context-dependent and should be interpreted with caution, as their general applicability across populations and deployment scenarios remains to be validated. The arc-cosine kernel (AK) emerged as an effective tool, particularly for key physiological traits with more complex genetic architectures, whereas the Gaussian kernel (GK) provided modest improvements for some traits but was generally less effective than AK. The linear G kernel remained competitive for core productivity

traits, such as growth and wood quality, but showed lower performance for chemical-defense and climate-response traits. These results also show that kernel choice, linear vs. non-linear, may influence ssGBLUP predictive performance in a trait-specific manner. Although some limitations were observed in multi-trait settings due to kernel averaging, our results suggest the potential of kernel-based approaches for capturing dominance and epistatic effects that linear models fail to represent. Finally, we highlight that adopting non-linear kernels, particular AK, may contribute to improved genomic prediction performance for complex phenotypes, to deliver the promises in accelerating breeding cycles, increasing genetic gain, and developing climate-resilience populations. To fully exploit the advantages of non-linear kernels in forest genetic improvement, future research should consider integrating machine learning kernels into multi-environment ssGBLUP frameworks and assess their capacity to capture genotype $\times$ environment interactions across diverse trials under changing climate conditions.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11295-026-01746-9>.

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Author contributions Eduardo P. Cappa: Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft, Writing - Review and Editing. Charles Chen: Data Curation, Resources, Writing - Review and Editing. Andy Benowicz: Data Curation, Resources, Funding acquisition, Writing - Review and Editing. Barb R. Thomas: Project Administration, Resources, Supervision, Validation, Writing - Review and Editing. Yousry A. El-Kassaby: Project Administration, Supervision, Validation, Writing - Review and Editing.

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Data availability https://github.com/RESFOR/quantitative_genetics_R/blob/main/White_Spruce_Pedigree_Phenotype_TGG2026.txt <https://github.com/REuantitGenotyping-by-sequencing> (GBS) raw reads used in this study have been deposited in NCBI SRA BioProject - PRJNA748443 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA748443>). Information of the white spruce Calling Lake trial including pedigree and adjusted phenotypic data are available in the https://github.com/RESFOR/quantitative_genetics_R/blob/main/White_Spruce_Pedigree_Phenotype_TGG2026.txt.

Declarations

Competing interests The authors declare no competing interests.

Data archiving Genotyping-by-sequencing (GBS) raw reads used in this study have been deposited in NCBI SRA BioProject - PRJNA748443 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA748443>) . Information of the white spruce Calling Lake trial including pedigree and adjusted phenotypic data are available in the GitHub repository: https://github.com/RESFOR/quantitative_genetics_R/blob/main/White_Spruce_Pedigree_Phenotype_TGG2026.txt.

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